

Saudi Mais Co, for Medical Products
Work Instruction

Title

Production & Test Procedure for Sterile Blood Lines

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1. Purpose:

The stipulate the production and testing procedure for all types of Sterile blood lines to ensure good quality products before releasing to the market.

2. Scope:

This procedure is applicable for all types of sterile blood line manufacturing by Saudi Mais Company.

3. Definition:

Bloodline used for conveying blood to and from a patient via a dialyzer, comprising: at least one assembly of tubing defining a flow path for the blood; a patient inflow end for receiving a flow of blood from the patient; a patient outflow end for returning the flow of blood to the patient.

4. Responsibilities:

- 4.1 Quality controllers are responsible for sampling and testing.
- 4.2 Production supervisors are responsible to produce good quality products.
- 4.3 Machine operators and assembly peoples are responsible to produced good quantity and good quality products.

5. Procedure:

5.1 Receiving Raw material:

- Ensure the received material from supplier is same as mentioned in purchase order.
- Inspect the material bag condition, must be no damaged at all bags.
- Take samples & inspect in order to AQL procedure, material description, specification with apply the AQL sampling plans.
- OBM inspector is responsible to submit material sample to analytical lab for testing material hardness and density and results should be match as per COA provided by supplier.
- Record into OBM inspection report.

5.1.1 Inspection & test procedure for Raw Material:

Work Description	Requirements	Method of Check
Clarity / Appearance	Clear / Good Appearance	Visual
Condition of Bag	Good / No damage bags	Visual
Material Grades / Description	As per P. O	Visual / Physical
Appearance of granules	Good / Transparent / Colored	Visual

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Granule's formation	No lobes formation	Visual
Lot number printed status	Clear printed	Visual
Arrangement of materials bags	Should be specified clear label	Visual

5.2. Injection Molding Machine Procedure:

- Production controller is responsible to issue required raw material for female connector, small connector blue and end plug.
- Production controller is responsible to make sure that the received material is same as per issue slip.
- Production controller is responsible to issue production order for products parts.
- Maintenance department is responsible to set the required mold on specified machine and set the standard parameters.
- Quality controller is responsible to verify the parameters and make sure that the required standard parameters are set.
- Machine operator is responsible to fill the material into machine hopper by using vacuum pump and fill hoppers with material up to the required capacity.
- Quality controller will take sample as per AQL general inspection method for normal inspection and AQL special inspection method for product performance testing all collect samples minimum cavities 2 to 3 shots piece/piece, if there is no any problem then machine run.

5.2.1 Female connector

Quality controller make sure that connector should be free from flashing, black spots, with proper thread and from damaged and crack.

5.2.2 Small Clamp Blue

Quality controller make sure that clamp should be free from flashing, black spots, good clamping and free from damaged and crack.

5.2.3 End Plug

Quality controller make sure that end plug should be free from flashing, black spots, and free from damaged and crack.

- Quality controller is responsible to check the products randomly every 2 hours & record the results in process control form.
- Machine operator is responsible to arrange good products in clean polyethylene bag.

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- Production controller are responsible to weigh the products and make inspection log of products then the QC must be sample on inspection log to be confirm acceptable before transfer to store or final process.

5.2 Extrusion Machine Procedure

Tools/Process

- 1. Chamber Tank
- 2. Pump System Tank
- 3. Repair Tank
- 4. Taping for Transport

- Production controller is responsible to issue raw material for extrusion tube and record the on after received raw material.
- Maintenance technicians are responsible to prepare required the for extrusion tube and record the on specified machine by using standard parameters.

- Machine operators are responsible to feed the material by manual feeding.
- Machine operators are responsible to submit initial sample to quality controller for inspection.

5.3.1 Visual Line machine inspection

- Quality controller will take sample as per AQL general inspection method for visual inspection and AQL special inspection method for performance testing and make sure that tubing should be smooth surface, leak resistance, transparent, accurate in dimension in ID, OD and length, straight cutting edges, free from black spots and burn marks.

- Every 2 hours QC will take samples from produced quantity base on AQL samples and check as per specification.

- Machine operator is responsible to arrange the good products in clean transparent polyethylene bags, make inspection tag and apply on the bag and QC will approve.

- Production controller is responsible to arrange the good products in specified rack or transfer to main components storage facility.

6. Receiving Bulk Components

- Ensure that received bulk components are same as mentioned in purchase order.
- Inspect the entire condition, must be in good condition, no damaged at all cartons with products.

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- Take samples & inspect in order to AQL procedure, and make sure received product are as per required specification.

6.1 Chamber 3way

Quality controller will make sure that 3way chamber should be transparent, free from black spots and should not damage crack and leakage.

6.2 Rotating Male connector Blue

Quality controller will make sure that rotating male connector should be free from black spots and should not damage crack and leakage.

6.3 Injection Port

Injection port should be clear and free from damaged and crack and without blockage.

6.4 Dialyzer Connector and cap

Dialyzer connector should be clear in blue or red color free from flashing back spots and blockage.

6.5 Big Clamp (Red & Blue)

Make sure both clamps should be smooth surface free from flashing and black spots and should be free from damaged and crack.

6.6 Transducer Filter.

Transducer filter body should be clear and transparent, proper sealed, no flashing, and free from damaged and crack.

6.7 Spike and protector

Quality controller is responsible to make sure that spike should be sharp tip, no tip damage and crack, and should be free from blockage.

6.8 Pump segment Connectors

Quality controller responsible to make sure that pump segment connectors should be clear and transparent, smooth surface and should be free from black spots, dust particles, fibers and irregularities.

6.9 T injection Site

Quality controller is responsible to make sure that T injection site rubber should be proper inserted and should be free from black spots, dust particles and free from damage and crack.

6.10 Recirculating Hook double female ports

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Quality controller make sure that recirculating hook double female ports good fitting with Male luer lock, no damaged, no crack, no leakage.

- Record the results into OBM inspection report.

7. Assembly Procedure:

7.1 Production controller will issue all components for assembly of arterial and venous blood line.

7.2 Production controller will provide approved drawing to assembled the components of blood lines, the approved drawing all types of blood lines referred to TF-21-PDBM. (Product drawing and bill of material of blood line)

7.3 Cut the tubing in accurate dimension as per mentioned in drawing and make sure that tube cutting surface should be straight and smooth and free from flashing and fibers.

7.4 Dip the tubing in cyclohexanone glue and insert into the relevant components as per drawing and make sure

the tubing should be proper assembled and no excess glue outside the connection tube.

7.5 The arterial and venous blood line shall be winding appropriately, in such a way that, when pack they are able to be place correctly or fit the package as good winding.

7.6 Every 2 hours QC will take samples from produced quantity base on AQL general inspection method for visual inspection and AQL special inspection method for performance testing.

7.7 Quality controller is responsible to check the products randomly every 2 hours & record the results in process control form.

7.6.1 QC in Process Control for Blood line:

(Arterial Line)

Sr.	Description	Requirements	Method of check
1	Clarity and Appearance	Clear and good appearance, no black spot, no flashing, no holes and free from dust particle.	Visual
2	End Plug	Smooth surface, no flashing, good fitting with FLL connector, no pullout.	Visual
3	Female connector	Connector should be transparent, no crack, no damaged, good fitting with end plug.	Visual
4	Clamps (Red) Status	Clamp should be smooth surface, no flashing, no crack, no damaged, no leakage.	Visual
5	T-injection Site	Strong gluing with tubes no pullout, no crack, no damaged, no leakage, red color, with rubber material good fitting inside port.	Visual / Physical
6	Transducer upper and	smooth surface, no crack no damaged, no leakage, proper	Visual /

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	lower part	sealed upper and lower part, strong gluing with tube, no pullout.	Physical
7	Transducer Tubing	Soft and kink resistance tubes, accurate in dimension, strong gluing with transducer filter and pump segment connector, no pullout.	Visual / Physical
8	Pump Segment Connectors	Smooth surface, accurate in dimension, no crack, no damaged, no leakage, good fitting with pump segment tube, no pullout.	Visual / Physical
9	Pump Segment Tube	Soft and kink resistance flexible tubes, accurate in dimension, strong gluing with connectors, no damaged, no holes, no leakage.	Visual / Physical
10	Heparin Tube Status	Should be clear and transparent, no black spots, accurate in dimension, good fitting with FLL connector, no leakage.	Visual / Physical
11	Chamber status	Clear and transparent PVC 2way chamber, proper fitting with chamber connector, strong gluing of tubing with chamber, no pullout, no leakage and free from black spot and dust particle.	Visual / Physical
12	Chamber Tube	Tube should be transparent, kink resistance, accurate in dimension, free from black spots, blockage and leakage	Visual / Physical
13	Dialyzer Connector (Red)	Strong gluing with tube, no pullout, no damaged, no leakage, no flashing and black spots.	Visual / Physical
14	Dialyzer connector cap	Should be good fitting with dialyzer connector, no damaged and crack.	Visual / Physical
15	Chamber tube with Transducer Filter	Clear and transparent kink resistance tube, good fitting with transducer, good fitting, strong gluing, no pullout, no crack and damaged.	Visual / Physical
16	FLL connector and cap status	FLL connector should be strong gluing with tube, no pullout, no crack, no damaged, cap good fitting with FLL connector, no leakage	Visual / Physical
17	Spike with protector Status	No damaged, no crack, no blockage, should be sharp tip, good fitting with protector.	Visual
18	Male luer lock connectors	Transparent connectors, good fitting, no cracks, no damaged, no leakage.	Visual
19	Leakage test	Applied pressure (38-42 Kpa) / (380 – 420mbar) / (5.5 – 6.2 Psi) Charge time 15sec, Testing time 10sec and should be free from leakage.	Visual / Physical
20	Winding Status	Should not bend, no blockage	Visual
Venous Line			
21	Clarity and Appearance	Clear and good appearance, no black spot, no flashing, no holes and free from dust particle	Visual
22	End Plug	Smooth surface, no flashing, good fitting with FLL connector, no pullout	Visual
23	Female connector	Connector should be transparent, no crack, no damaged, good fitting with end plug	Visual

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24	Clamps (Blue) Status	Clamp should be smooth surface, no flashing, no crack, no damaged, no leakage.	Visual
25	T-injection Site	Strong gluing with tubes no pullout, no crack, no damaged, no leakage, blue Colour, with rubber material good fitting inside port.	Visual / Physical
26	Chamber tube with dialyzer Connector & cap	Tube good and strong fitting with dialyzer connector and T injection site, strong gluing, no pullout, no leakage between tube and connector, accurate in dimension, cap good fitting with connector.	Visual / Physical
27	Transducer upper and lower part (monitor line)	smooth surface, no crack no damaged, no leakage, proper sealed upper and lower part, strong gluing with tube, no pullout	Visual
28	Transducer Tubing with chamber (monitor line)	soft and kink resistance tubes, accurate in dimension, strong gluing with transducer filter and chamber, no pullout.	Visual / Physical
29	Chamber tube with FLL connector status (monitor line)	Tube good and strong gluing with chamber and transducer filter, proper sealed, transducer filter no pull out from tubing, strong gluing, no leakage, good fitting of end plug and FLL connector	Visual / Physical
30	Chamber status	Clear and transparent PVC 3way chamber, proper fitting with chamber connector, strong gluing of tubing with chamber, no pullout, no leakage, and free from black spot and dust particle.	Visual / Physical
31	Chamber filter Status	Good fitting inside chamber, no blockage, and should be free from damaged and crack	Visual / Physical
32	Level Adjustment line Status	Clear and transparent tube, accurate in dimension, good fitting with chamber by strong gluing, no pull out, no leakage and blockage.	Visual / Physical
33	Spike with protector Status	No damaged, no crack, no blockage, should be sharp tip, good fitting with protector	Visual
34	Male luer lock connectors	Transparent connectors, good fitting, no cracks, no damaged, no leakage	Visual / Physical
35	Recirculating Hook double female ports Status	Good fitting with Male luer lock, no damaged, no crack, no leakage	Physical
36	Leakage test	Applied pressure (38-42 Kpa) / (380 – 420mbar) / (5.5 – 6.2 Psi) Charge time 15sec, Testing time 10sec and should be free from leakage.	Visual / Physical
37	Winding Status	Should not bend, no blockage	Visual

8. Leak Testing process.

- Each blood line arterial and venous to be check one by one to make sure that the assembled blood line leakage free.

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- Operator is responsible to make sure that the leak tester approved parameters charge time and duration time proper set as mentioned in given table.

Pressure Setting For Blood Line	Charge Time (± 2 sec)	Test Duration Time (± 2 sec)
38 - 42 Kpa	15 sec.	10 sec.
380 - 420 mbar	15 sec.	10 sec.
5.5 - 6.2 psi	15 sec.	10 sec.

- Connect arterial line with leak test nozzle and press "ON" button of leak tester, then observe it until test duration time complete.
- Repeat the process for venous line leak test.
- The Green light indication acceptable and if red light indication rejected.
- Quality controller is responsible to verify parameters for blood line leak test.
- Operators are responsible to arrange good quantity in clean poly bag and poly bag should be with inspection tag and approved by QC.
- Production controller is responsible to arrange good quantity in specified area or transfer to next process.

9. Packaging and Labeling:

- Production controller is responsible to issue packaging film and paper as per approved packaging film and paper specification and issue production order for packaging.
- Machine operator is responsible to set the required packaging machine die and set approved standard parameters.
- Machine operator is responsible to set the approved clichés and make sure that clichés information are correct as per packaging order.
- Quality controller is responsible to the check, Product name and code no., reference no., size, lot no., type, temperature logo, sterilization logos, production date, expiry date, package forming, printing, sealing and cutting etc. and must be stamp on pouch as approved.
- After packaging on machine, the product should be checked by the repacking personnel, packaging should be intact & in good condition.
- Then pack in specified small boxes, and arrange in big carton.
- The big carton must be with label and detail information.

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10. Sterilization:

- The finished products are sterilized as per work instruction WI-7-5-1-1.

11. Final Inspection and Product Release:

- The Quality controller must be check, the unloaded products after sterilization.
- Quality controller is responsible to take sample as per AQL general inspection method for visual inspection and AQL special inspection method for performance testing of product and inspect the products as requirements mention in final inspection and release.
- The quality assurance manager or quality control manager have to review the completed cycle parameters and inspection reports both documents have to be sign.
- After confirmation all the tests from the Lab, to release the sterilized products, the document once again have to review the quality control manager then sign.
- The production department responsible to record the quantity of products before transferring to the finished goods ware-house and then it could be released to the customer.

12. Sampling:

- Inspection & Test Report for Raw Material : F-823-T1
- Inspection & Test Report for Blood line parts : F-823-T76
- In-process control for Blood Line assembly : F-823-i16
- In-process control for Packaging machine and final checking : F-823-i4
- Final Inspection and Release Record for blood line : F-823-F6

13. Reference:

BS EN ISO 8637-2 Extracorporeal Blood Circuits for Haemodialysers, Haemodiafilters and haemofilters.

14. Modification / Documentation Change History:

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22.11.2018	Item 2,3, 4.3, 7,8,9 updated	3
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01.09.2023	Clause 5,6,7,8,9,11 and 12 added	5

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